MATH 350-2 Advanced Calculus

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Outline

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Adherent and Accumulation Points

Definition

A point $\vec{x} \in \mathbb{R}^n$ is called an **adherent point** if for all $r > 0$ the ball $B(\vec{x}; r)$ contains at least one element of A.

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- -1 is an accumulation point of $(-1, 1)$.
- suprema and infima are accumulation points!
- 0 is an accumulation point of $\{1/1, 1/2, 1/3, \ldots\}$

Characterizing accumulation points

Theorem (Apostol Theorem 3.17)

A point $\vec{x}$ is an accumulation point of A if and only if for all $r > 0$, the ball $B(\vec{x}; r)$ contains infinitely many points of A.

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Obviously, if for all $r > 0$, the ball $B(\vec{x}; r)$ contains infinitely many points of A, then it contains at least one point of A different from $\vec{x}$, so it's an accumulation point.

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Suppose instead that $\vec{x}$ is an accumulation point and let $r > 0$.

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Suppose instead that $\vec{x}$ is an accumulation point and let $r > 0$. It suffices to show that the set $C = (B(\vec{x}; r)\backslash\{\vec{x}\}) \cap A$ is infinite. $\square$

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Suppose that C is finite.

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Suppose that C is finite.
Then the set $\{|\vec{y} - \vec{x}| : \vec{y} \in C\}$ is also finite.

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It is also nonempty and has only positive values.

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However, $\vec{y} \in C$ and $|\vec{y} - \vec{x}| < s$, contradicting the minimality of s.

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It is also nonempty and has only positive values. This means that it has a minimum value $s > 0$.
Since $\vec{x}$ is an accumulation point, $B(\vec{x}; s) \cap A$ has an element $\vec{y}$ different from $\vec{x}$.
However, $\vec{y} \in C$ and $|\vec{y} - \vec{x}| < s$, contradicting the minimality of s.
We conclude that C is infinite. $\square$

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Definition

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Theorem (Apostol Theorem 3.18, 3.20, 3.22)

A set is closed if and only if it contains all of its adherent points (ie. $A = \overline{A}$), or equivalently if and only if A contains all of its accumulation points.

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If $\vec{x} \in \mathbb{R}^n \backslash A$, then there exist $r > 0$ such that $B(\vec{x}; r) \subseteq \mathbb{R}^n \backslash A$.

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Thus $\vec{x}$ is not an adherent point of A.

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Thus $\vec{x}$ is not an adherent point of A.
Thus every adherent point of A is an element of A.

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Conversely, suppse that A contains all of its accumulation points.

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Since $\vec{x}$ was an arbitrary element of $\mathbb{R}^n \backslash A$, this proves $\mathbb{R}^n \backslash A$ is open.

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Closures are closed

Theorem

If $A \subseteq \mathbb{R}^n$ is any set, then $\overline{A}$ is closed.

Proof.

Exercise! $\square$

Outline

1. Real Analysis Lecture 7
   - More on Closed Sets
   - Compactness

Warm-up Challenge

Problem

Show that the closed ball

$$\overline{B}(\vec{x}; r) = \{\vec{y} \in \mathbb{R}^n : |\vec{x} - \vec{y}| \leq r\}$$

is a closed set.

Point set topology

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Answers must be in terms of open sets

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- $\{\mathbb{R}\}$ is an open cover of the interval $(0, 1]$
- $\{(0, 1/2), (1/2, 2)\}$ is an open cover of the interval $(0, 1]$
- $\{(\frac{1}{n}, \frac{n+1}{n}) : n \in \mathbb{Z}_+\}$ is an open cover of $(0, 1]$

Lindelöf Covering Theorem

Theorem (Lindelöf Covering Theorem)

Let $A \subseteq \mathbb{R}^n$ be a set and suppose $\{U_i : i \in I\}$ is an open covering of A. Then there exists a countable subcover $\{U_j : j \in J\}$.

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Can we do better than this?

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Question

Can we do better than this?

Definition

A set $A \subseteq \mathbb{R}^n$ is called **compact** if every open cover of A has a *finite* subcover.

Challenge

Problem

Show that a singleton set $A = \{\vec{x}\}$ is compact.

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Suppose that $\{U_i : i \in I\}$ is an open cover of X.
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It follows that $\{U_j\}$ is a subcover of A consisting of a single set.

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Therefore $\vec{x} \in U_j$ for some $j \in I$.
It follows that $\{U_j\}$ is a subcover of A consisting of a single set.
Thus every open cover has a finite subcover, making A compact!

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Proof.

Suppose A is a compact set.
Consider the family of sets $\{U_i : i \in I\}$ where $I = \mathbb{Z}_+$ and $U_i = B(\vec{0}; i)$.

$\square$

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Note that U_i is an open subset for all i, so we need only show $A \subseteq \bigcup_{i\in I} U_i$.

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$$\bigcup_{i \in I} U_i = \bigcup_{i=1}^{\infty} B(\vec{0}; i) \subseteq \mathbb{R}^n.$$

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Solution

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Thus $\mathbb{R}^n = \bigcup_{i \in I} U_i$, and since $A \subseteq \mathbb{R}^n$, we see $A \subseteq \bigcup_{i \in I} U_i$.

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There is a finite subset $\{i_1, i_2, \ldots, i_m\} \subseteq I$ with $A \subseteq \bigcup_{k=1}^m U_{i_k}$.

Challenge

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Show that $\bigcup_{k=1}^{m} U_{i_k}$ is an open ball, and thus A is bounded.

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This shows A is bounded.

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Theorem

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Consider the family of sets $\{U_i : i \in I\}$ where $I = \mathbb{Z}_+$ and $U_i = \mathbb{R}^n \backslash \overline{B}(\vec{x}; 1/i)$. □

Challenge

Problem

Prove that $\bigcup_{i \in I} U_i = \mathbb{R}^n \backslash \{\vec{x}\}$.

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Problem

Explain why $\{U_i : i \in I\}$ is an open cover of A.

Challenge

Problem

Show that if $\{U_i : i \in I\}$ has a finite subcover, then $\vec{x}$ can't be an adherent point of $\vec{A}$.

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Theorem (Bolzano-Weierstrass Theorem)

A bounded set $A \subseteq \mathbb{R}^n$ with infinitely many points must have an accumulation point.

Cantor Intersection Theorem

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Suppose that $C_1, C_2, C_3, \dots$ are non-empty closed, bounded sets with $C_{i+1} \subseteq C_i$ for all $i \in I = \mathbb{Z}_+$. Then $\bigcap_{i \in I} C_i$ is also non-empty.

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Therefore the Bolzano-Weierstrass Theorem tells us it has an accumulation point $\vec{x}$.

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Since C_m is closed, it follows $\vec{x} \in C_m$, and thus $\vec{x} \in \bigcap_{i \in I} C_i$. In particular the intersection is non-empty! □

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This is a contradiction. $\square$